

Auto Data Engineering Pipeline

Making DCAT-AP Datasets AI-Ready

Currently Data Platform lacks AI-Readiness

- Most datasets
 - uncleaned
 - unstandardized
 - have no defined task -> have no defined train / test splits
- -> not possible to use for AI out-of-the-box
- We need dataset-specific processing scripts
 - function that takes in path to raw data -> outputs dataframes for train test

Idea: Automate with Agentic AI

- Input
 - datasets metadata (DCAT-AP) and raw data
 - instruction including output format
 - isolated sandbox environment
- Agent should
 1. analyze metadata and data -> define task
 2. analyze raw data -> write processing script constrained interface
 3. test processing script
 4. write ml croissant file
- Output: processing script, ml croissant file, logs

Resources

- <https://pypi.org/project/dcat-ap-hub/>
- <https://ki-daten.hlr.de/de/dataset/10-5281-zenodo-13808085>
- <https://lmstudio.ai/>
- <https://ollama.com/>
- <https://github.com/Chachamaru127/claude-code-harness>
- <https://www.langchain.com/langgraph>
- <https://www.zml.kit.edu/english/ki-toolbox-en.php>

Pipeline (Later)

1. Discovery

1. Connect to piveau hub REST API
2. Filter out datasets that are not
3. Put them in queue

For each item in the queue:

1. Retrieval

1. download metadata and raw data with dcat-ap hub

2. Data Engineering via Agent -> py script (data processor), train.csv, test.csv, croissant.jsonld

2. Provide mlcoirssant and processing cscript to piveau hub